

✓ **Project** | Sustainability Impact Analysis for Intel



INTRODUCTION: As you learned listening in on the strategy meeting with Dr. Alvarez and Intel's Sustainability Team, Intel is committed to reducing its carbon footprint and improving the sustainability of its devices – not just during manufacturing, but throughout the entire lifecycle.

A key part of this effort is their repurposing programs, which play a central role in achieving these sustainability goals. Repurposing and recycling programs aim to reduce e-waste, energy consumption, and CO₂ emissions by extending the life of existing devices, and thus reducing the need for new device manufacturing. Like Michael Campbell said: the average household in the US has anywhere from 3–5 PCs devices, tablets, notebooks, desktops that are perfectly functional, but not being used!

One challenge Intel faces is determining which devices in its repurposing program should be prioritized for the maximum environmental benefit. That's where data analysis comes in! To help with this, Intel gathered data on each device repurposed or recycled in 2024.

Your task is to evaluate the effectiveness of Intel's current repurposing strategy and provide a data-driven recommendation to help guide the program's direction and optimize sustainability efforts.

HOW IT WORKS: Follow the prompts in the questions below to investigate the data. Post your answers in the provided boxes: the **yellow boxes** for the queries you write and **blue boxes** for your text-based analysis. Once you're done, you'll submit your **completed** .pdf file to HQ for feedback from The Accelerator Team.

SQL App: [Here's the link](#) to our specialized SQL app, where you'll write your SQL queries and interact with the data.

NOTE: The dataset you are working with is designed for The Global Career Accelerator to reflect the key characteristics and structure of Intel's real data, while protecting their confidentiality and proprietary information. Be aware that any conclusions or results derived from this dataset should be viewed as hypothetical and for illustrative purposes only.

– Data Set **Descriptions**

In this project you'll query 2 different datasets, `intel.device_data` and `intel.impact_data`, that you will join together for your analysis. Here you'll find the data dictionary for each dataset.

`intel.device_data`

- `device_id`: Unique identifier for each repurposed device
- `device_type`: Type of device, values are either "Laptop" or "Desktop"
- `model_year`: The year the device was manufactured (e.g., 2018, 2019, etc.)

`intel.impact_data`

- `impact_id`: Unique identifier for the repurposed device's impact record (e.g., "LP20NA141592")
- `device_id`: Unique identifier linking the impact record to a specific device in the `intel.device_data` table
- `usage_purpose`: The specific purpose for which the device is being repurposed, values are Education & Digital Literacy, Corporate & Enterprise, Government & Public Sector, Environmental Sustainability Programs, and Social Impact & Non-Profit
- `power_consumption`: Power consumption of the device in watts (W) when in use (e.g., 50W, 75W)
- `energy_savings_yr`: Estimated energy savings per device per year when repurposed compared to a new device, measured in kilowatt-hours (kWh)
- `co2_saved_kg_yr`: Estimated CO2 emissions saved per device per year from manufacturing a new device, measured in kilograms (kg).
- `recycling_rate`: The percentage of the device that is recyclable (e.g., 80%, 90%).
- `region`: The geographical region where the device was repurposed, values are "North America", "Europe", and "Asia"

– **Task 1:** Organizing and Understanding the Data

We'll start by **joining** the device data with the impact data, allowing for a comprehensive analysis of device types, model years, repurpose regions, and energy savings in one dataset.

- A. Simply write a query that returns all of the columns from both tables, joining the two on the `device_id` column. Be sure to choose the appropriate join so that all relevant

data is included in your result. When done properly, the result should have 601,740 rows.

(paste your query below 📌)

```
SELECT*
FROM
  intel.device_data AS d
  JOIN intel.impact_data AS i ON d.device_id =
i.device_id
```

This query is limited to only the first 150000 rows,
but when I inserted a COUNT function it showed 601740.

- B.** To your joined dataset, add a new column called `device_age` calculated by subtracting the `model_year` from 2024. Paste your query below and double check that the values in your new column make sense. For example, a 2019 device should be 5 years old.

(paste your query below 📌)

```
SELECT
  2024 - d.model_year AS device_age,
  *
FROM
  intel.device_data AS d
  JOIN intel.impact_data AS i ON d.device_id =
i.device_id
SELECT
  *,
  2024 - d.model_year AS device_age
FROM
  intel.device_data AS d
  JOIN intel.impact_data AS i ON d.device_id =
i.device_id
```

- C. Order your joined data by `model_year` (oldest to newest). Do you notice more older (5+ years) or newer (under 5 years) devices being repurposed? What might that indicate?

(write your **answer** below 📌)

The number of newer devices being repurposed is higher. This likely represents a broader trend of users prioritizing having the newest upgrade over using their devices until failure.

- D. Bucketing the `device_age` will allow us to analyze trends and patterns in energy savings and CO₂ reductions more effectively than using individual ages. Use a `CASE WHEN` clause to add one more column, called `device_age_bucket`, to your data, that is based on the `device_age`:

- `WHEN` the `device_age` is less than or equal to 3, `device_age_bucket` should be “newer”
- `WHEN` the `device_age` is greater than 3 but less than or equal to 6, `device_age_bucket` should be “mid-age”
- `WHEN` the `device_age` is greater than 6, `device_age_bucket` should be “older”

HINT: Instead of using e.g. `device_age <= 3`, you need to reference the calculation directly: `2024 - d.model_year <= 3`.

Double check that the values in your new column make sense! For example, a 2019 device should be characterized as “mid-age”.

(paste your query below 📌)

```
SELECT
  2024 - d.model_year AS device_age,
  COUNT(*) AS num_devices,
  CASE
    WHEN 2024 - d.model_year <= 3 THEN 'newer'
    WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
```

```
        ELSE 'older'
      END AS device_age_bucket
FROM
  intel.device_data AS d
  JOIN intel.impact_data AS i ON d.device_id =
    i.device_id
GROUP BY device_age_bucket, device_age
ORDER BY device_age DESC
```

– Task 2: Key Insights

Now it's time to analyze the overall impact of Intel's repurposing program. You will use your final query from **Task 1** together with the **WITH** keyword for the remainder of this Project as you aggregate and analyze the data you've organized and prepped. For a refresher, rewatch “ The **WITH** Keyword” in SkillBuilder 6.

A. What is the total number of devices Intel repurposed in 2024?

(write your **answer** below )

601740

B. Write a query that returns the total number of devices repurposed, the average age of repurposed devices in 2024, the average estimated energy savings (kWh) from repurposed devices per year, and the total CO₂ emissions saved (in tons) from repurposed devices.

Note: CO₂ emissions are typically measured in tons. Since **CO₂_saved_kg_yr** is measured in kg, divide the **SUM(CO₂_saved_kg_yr)** by 1000 to report the total CO₂ emissions saved in tons.

(paste your query below )

```
SELECT
  COUNT(*) AS num_devices,
```

```
AVG(2024 - d.model_year) AS avg_age,  
AVG (energy_savings_yr) AS avg_nrg_savings,  
SUM(i.co2_saved_kg_yr) / 1000 AS total_co2_saved_tons  
FROM  
intel.device_data AS d  
JOIN intel.impact_data AS i  
ON d.device_id = i.device_id
```

- C. Now that you have calculated the average estimated energy savings (kWh) and CO₂ emissions saved (tons), use ChatGPT to help put these numbers into perspective.



Try this prompt: I found that each repurposed device saves approximately of XXX kWh of energy per year and Intel's repurposing program saved XXX tons of CO₂ emissions in one year. Help me understand the significance of these numbers. How would this compare to the energy consumption of a small city or the amount of CO₂ produced by cars? What is the environmental impact of these savings?

What comparisons did you find most impactful in terms of scale? Summarize how much energy and CO₂ emissions were saved and how it compares to something familiar, like powering households or reducing car emissions.

(write your **answer** below 🖊)

That's enough energy to power a small town for a year and enough emissions to take almost 1500 cars off the road for a year.

– Task 3: Identifying Trends & Maximizing Sustainability

By grouping our data in different ways, we can uncover patterns in energy savings and CO₂ reductions. These insights will help us determine which categories of devices contribute the most to sustainability efforts and where Intel should focus its repurposing strategy for maximum impact.

- A. Write a query that returns the total number of devices, the average energy savings, and the average CO₂ emissions saved (in tons), grouped by device_type.

Note (again): You'll need to divide `AVG(CO2_saved_kg_yr)` by 1000 to report the average CO₂ emissions saved in tons.

(paste your query below 📌)

```
SELECT
  d.device_type,
  COUNT(d.device_id) AS num_devices,
  AVG(2024 - d.model_year) AS avg_age,
  AVG (energy_savings_yr) AS avg_nrg_savings,
  SUM(i.co2_saved_kg_yr) / 1000 AS total_co2_saved_tons
FROM
  intel.device_data AS d
  JOIN intel.impact_data AS i ON d.device_id =
i.device_id
GROUP BY
  d.device_type
```

- B. Based on the results, which device type contributes the most to energy savings and CO₂ reduction? Why might that be the case?

Hint: Don't forget you can use ChatGPT as your Teammate to help think through your response!

(write your **answer** below 📌)

Laptops are almost twice as productive than desktops when it comes to CO₂ emissions savings, but only slightly better for energy savings on average. Laptops save only a bit more energy than desktops because they're already efficient by design – but they prevent a lot more CO₂ when repurposed, since they avoid the need to manufacture another high-impact device.

- C. Write a query that returns the total number of devices, the average energy savings, and the average CO₂ emissions saved (in tons), now grouped by device_age_bucket.

(paste your query below 📌)

```
SELECT
  CASE
    WHEN 2024 - d.model_year <= 3 THEN 'newer'
    WHEN 2024 - d.model_year <= 6 THEN 'mid-age'
    ELSE 'older'
  END AS device_age_bucket,
  COUNT(*) AS num_devices,
  AVG(i.energy_savings_yr) AS avg_nrg_saved,
  AVG(i.co2_saved_kg_yr) / 1000 AS avg_co2_saved_tons
FROM
  intel.device_data AS d
JOIN
  intel.impact_data AS i
  ON d.device_id = i.device_id
GROUP BY
  device_age_bucket
ORDER BY
  num_devices DESC
```

- D. Based on the result of your query, what do you notice about the relationship between device age and the number of devices repurposed versus the average energy saved?

(write your **answer** below 📌)

Most devices being repurposed are newer, but the energy and CO₂ savings are significantly lower on average than older devices.

- E. Finally, write a query that returns the total number of devices, the average energy savings, and the average CO₂ emissions saved (in tons), now grouped by region.

(paste your query below 📌)

```
SELECT
  i.region,
  COUNT(*) AS num_devices,
  AVG(i.energy_savings_yr) AS avg_nrg_saved,
  AVG(i.co2_saved_kg_yr) / 1000 AS avg_co2_saved_tons
FROM
  intel.device_data AS d
JOIN
  intel.impact_data AS i
  ON d.device_id = i.device_id
GROUP BY
  i.region
ORDER BY
  num_devices DESC
```

- F. How does the carbon intensity of electricity in each region impact the total CO₂ savings from repurposed devices? Are there regions where repurposing leads to significantly higher environmental benefits? Why might that be?

(write your **answer** below 📌)

I don't think I understand what this question is asking. I don't see any correlation between energy and CO₂ savings, but Europe has significantly lower average CO₂ savings. Asia has the second highest number of devices, but the highest average CO₂ savings– so that seems like more environmental benefits for the same process.

– Task 4: Data-Driven Recommendations

Using the findings from this analysis, we need to summarize key takeaways and develop actionable recommendations for Intel. Remember: the goal is to refine Intel's repurposing strategy to maximize energy savings and CO₂ reductions while ensuring the most effective use of resources.

- A.** Based on your analysis of the repurposed devices (including energy savings, CO₂ emissions, and device age), write **four** key takeaways in succinct sentences/bullets that summarize the most important patterns and insights from the data. These should be specific, concise, and focused on the implications of repurposing newer versus older devices.

(write your **answer** below 🖊)

Some countries, like Asia, will benefit more from repurposing their devices.

North American consumers are consuming more devices than other countries.

Most devices are newer, which likely means customers are prioritizing having the newest device over keeping their older one.

Repurposing newer devices does offer some savings, but not as much as repurposing older devices.

- B.** Based on your four key takeaways and ChatGPT as your teammate, write a recommendation for Intel on how to improve the repurposing program. Your recommendation should include a clear action or strategy for Intel based on the data and a data-driven justification for why this approach would maximize energy savings and CO₂ reductions.

(write your **answer** below 🖊)

To enhance its device repurposing program, Intel should implement a geographically tailored strategy: bolstering collection efforts in high-consumption markets like North America while expanding repurposing initiatives in regions like Asia, where the benefits are more pronounced. Recognizing the trend of frequent upgrades, particularly for newer devices, requires both consumer education on environmental impact and a focus on component recovery, since the direct financial savings from repurposing these newer models are less significant compared to older tech, which should continue to be prioritized for full refurbishment. This dual approach aims to maximize both economic and environmental returns.

- C. Briefly reflect on how ChatGPT's suggestions influenced your recommendation. Did it help you see something you hadn't considered? What parts of your recommendation were improved based on its response?

(write your **answer** below 📌)

Since our data focused solely on the environmental aspect, I hadn't thought about the financial benefit of this program. Repurposing old devices to create new tech will reduce emission, energy costs, and financial burden as well.

– LevelUp: Optimizing Repurposing Strategy for Maximum Impact

Now that you've gained insights into the energy savings and CO₂ reductions across different device types and regions, let's use this data to optimize Intel's repurposing strategy for maximum environmental benefit.

- A. Add to your final query of Task 3 that returns the total number of devices, the average energy savings, and the average CO₂ emissions saved (in tons), grouped by region, **the percentage** of the total energy savings and CO₂ reductions contributed by each device type within each region.

HINT: To calculate the percentage of the total energy savings, use this formula:

Total energy savings for the device type / Total energy savings for the region) * 100
You'll use a similar one for the percentage of the total CO₂ reductions.



Try this prompt: What's the best way to calculate the percentage of CO₂ reductions contributed by each device type in each region?

(paste your query below 📌)

```
SELECT  
  i.region,
```

```

d.device_type,
COUNT(*) AS num_devices,
AVG(i.energy_savings_yr) AS avg_energy_savings,
CASE
    WHEN SUM(i.energy_savings_yr) = 0 THEN 0.0
    ELSE COUNT(d.device_type) * 1.0 /
SUM(i.energy_savings_yr)
END AS avg_nrg_sav_by_dt,
(AVG(i.co2_saved_kg_yr) / 1000) AS
avg_co2_saved_tons,
(
    SUM(i.energy_savings_yr) * 100.0 /
SUM(SUM(i.energy_savings_yr)) OVER (PARTITION BY
i.region)
) AS pct_energy_savings_region,
(
    SUM(i.co2_saved_kg_yr) * 100.0 /
SUM(SUM(i.co2_saved_kg_yr)) OVER (PARTITION BY
i.region)
) AS pct_co2_saved_region
FROM
    intel.device_data AS d
JOIN intel.impact_data AS i ON d.device_id =
i.device_id
GROUP BY
    i.region,
    d.device_type
ORDER BY
    num_devices DESC

```

B. Based on the results of your query, analyze the data to answer:

- Which device types in which regions contribute the most energy savings and CO₂ reductions relative to their numbers?
- How can this analysis help Intel prioritize specific device types in certain regions to maximize environmental benefits?

(write your **answer** below 🖱)

Based on the analysis, Intel should prioritize its repurposing efforts by globally focusing on **Laptops**, as they consistently represent the highest volume of devices and thus contribute the most to overall regional energy and CO₂ savings. Crucially, Intel should strategically intensify its efforts in **Asia**, where both Laptop and Desktop devices demonstrate significantly higher *per-device* CO₂ reductions, indicating a greater environmental return on investment for each unit repurposed there. While maintaining high-volume collection in North America due to its sheer consumption, this targeted regional and device-type focus will allow Intel to maximize its environmental benefits from the repurposing program.

- C. In addition to focusing on sustainability, imagine Intel needs to optimize for cost-effectiveness in their repurposing program. How might you adjust your query to incorporate cost data (e.g., cost per repurposed device)? What strategies could Intel use to balance sustainability goals with cost constraints?

(write your **answer** below 🖊)

We don't have any cost data to examine in our dataset, but if we did, we could see how much these efforts are saving Intel. We would need to see how much it costs to repurpose these devices compared to manufacturing new ones. To balance sustainability with cost, Intel could prioritize repurposing efforts in regions and for device types that offer the highest environmental benefits (like Asia for CO₂ savings) at the lowest cost per unit of impact, while optimizing collection logistics in high-volume areas like North America to reduce per-device expenses. Furthermore, strategies like diversifying repurposing processes based on device age (e.g., component recovery for newer devices vs. full refurbishment for older, more cost-effective ones) and leveraging partnerships could help maximize the net positive impact.

